MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)

Third Semester Examination

(2024 Admission Onwards)

MG3DSCMAT201 Building Blocks for Higher Mathematics

Duration: 2 hrs **Maximum Marks: 70**

Students should attempt atleast one question from each course outcome to enhance their overall attainability.

PART A

Answer any five questions. Each question carries 2 marks.

1. Show that the proposition $P(0)$ is true, where $P(n)$ is "If $n > 1$, then $n^2 > n$" and the domain consists of all integers. [S][7]
2. How many relations are there on a set with n elements? [An][2]
3. What are the equivalence classes of 1 and 3 modulo 5? [U][3]
4. Is the poset $(Z^+, |)$ totally ordered? Why? [U][3]
5. Find the velocity and acceleration vectors of a particle whose motion in space is given by the position vector $\mathbf{r}(t) = (t+1)\,\mathbf{i} + (t^2 - 1)\,\mathbf{j}$ at t = 1. [A][4]
6. State and prove Dot Product Rule for differentiating vector functions. [A][4]
7. Find Curl F for the vector field $F = x^2 z\,\mathbf{i} - 2y^3 z^2\,\mathbf{j} + xy^2 z\,\mathbf{k}$. [An][5]
8. State Divergence theorem. [A][6]

PART B

Answer any five questions. Each question carries 6 marks.

9. Prove that a relation R on a set A is transitive if and only if $R^n \subseteq R$ for $n = 1, 2, 3, ...$ [An][2]

10. (a) Suppose that the relations R_1 and R_2 on a set A are represented by the matrices $M_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ and $M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$. What are the matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$? [An][2]

(b) Find the matrix representing the relation $S \circ R$, where the matrices representing R and S are $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$. [An][2]

11. (a) Define equivalence relation. Determine whether the relation represented by the zero-one matrix $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ is an equivalence relation. [U][3]

(b) Is the collection of subsets $\{1, 2\}, \{2, 3, 4\}, \{4, 5, 6\}$ a partition of the set $\{1, 2, 3, 4, 5, 6\}$? Justify. [U][3]

12. (a) Find the greatest lower bound and the least upper bound of the sets, $\{6, 12, 18\}$ and $\{1, 3, 5, 8\}$ if they exist, in the poset $(Z^+, |)$. [U][3]

(b)Determine whether the poset $(\{1, 5, 25, 125\}, |)$ is a lattice. [U][3]

13. Find the length of the curve $\mathbf{r(t)} = \sqrt{2}t\,\mathbf{i} + \sqrt{2}t\,\mathbf{j} + (1 - t^2)\,\mathbf{k}$ from $(0, 0, 1)$ to $(\sqrt{2}, \sqrt{2}, 0)$. [A][4]

14. Define directional derivative of a function in the plane. Using the definition, find the derivative of $f(x, y) = 2x^2 + y^2$ at $P_0(-1, 1)$ in the direction of $\mathbf{u} = 3\mathbf{i} + 4\mathbf{j}$. [A][4]

15. A fluid's velocity field is $\mathbf{F} = x\,\mathbf{i} + z\,\mathbf{j} + y\,\mathbf{k}$. Find the flow along the helix $\mathbf{r(t)} = (\cos t)\,\mathbf{i} + (sint)\,\mathbf{j} + t\,\mathbf{k}; 0 \leq t \leq \dfrac{\pi}{2}$ [An][5]

16. Show that $\mathbf{F} = (e^x \cos y + yz)\,\mathbf{i} + (xz - e^x \sin y)\,\mathbf{j} + (xy + z)\,\mathbf{k}$ is conservative over its natural domain and find a potential function for it. [An][5]

PART C

Answer any three questions. Each question carries 10 marks.

17. (a) Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd". [A][7]
(b)Prove that if n is an even integer and $3n + 2$ is odd, then n is odd by the contraposition method. [A][1]
(c) Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction. [A][1]

18. Let R be an equivalence relation on a set A and let $a, b \in A$. Show that the following three statements are equivalent:
i. aRb
ii. $[a] = [b]$
iii. $[a] \cap [b] \neq \varnothing$. [U][3]

19. Draw the Hasse Diagram for inclusion on the set $P(S)$, where $S = \{a, b, c, d\}$. [U][3]

20. (a)Find the curvature for the helix $\mathbf{r}(t) = a\cos t\,\mathbf{i} + a\sin t\,\mathbf{j} + bt\,\mathbf{k}$, $a, b \geq 0, \quad a^2 + b^2 \neq 0$. [A][4]
(b) Find $\mathbf{N}$ for the helix and describe how the vector is pointing.

[A][4]

21. Integrate $f(x, y, z) = x + \sqrt{y} - z^2$ over the path from $(0, 0, 0)$ to $(1, 1, 1)$ given by

(a) $C_1 : \mathbf{r(t)} = t\,\mathbf{i} + t^2\,\mathbf{j}$, $0 \leq t \leq 1$

(b) $C_2 : \mathbf{r(t)} = \mathbf{i} + \mathbf{j} + t\,\mathbf{k}$, $0 \leq t \leq 1$ [An][5]

22. Verify both forms of Green's theorem for the vector field $\mathbf{F} = (x - y)\,\mathbf{i} + x\,\mathbf{j}$ and the region R is bounded by the unit circle $C : \mathbf{r(t)} = (\cos t)\,\mathbf{i} + (\sin t)\,\mathbf{j}$, $0 \leq t \leq 2\pi$ [A][6]